Recursion Tree Method for Mixed Work Function

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Course: Design and Analysis of Algorithm

Problem Statement

Problem Statement

- Analyze the recurrence relation:
- **$T(n) = 3T(n/2) + n^2$**
- using the **Recursion Tree Method**.

Objectives:

- Construct the recursion tree.
- Compute the cost at each level.
- Sum all costs.
- Find the overall time complexity.

Algorithm Overview

Recursion Tree Method

The Recursion Tree Method:

- Expands the recurrence into a tree.
- Each node represents a recursive call.
- Calculates the work done at every level.

For

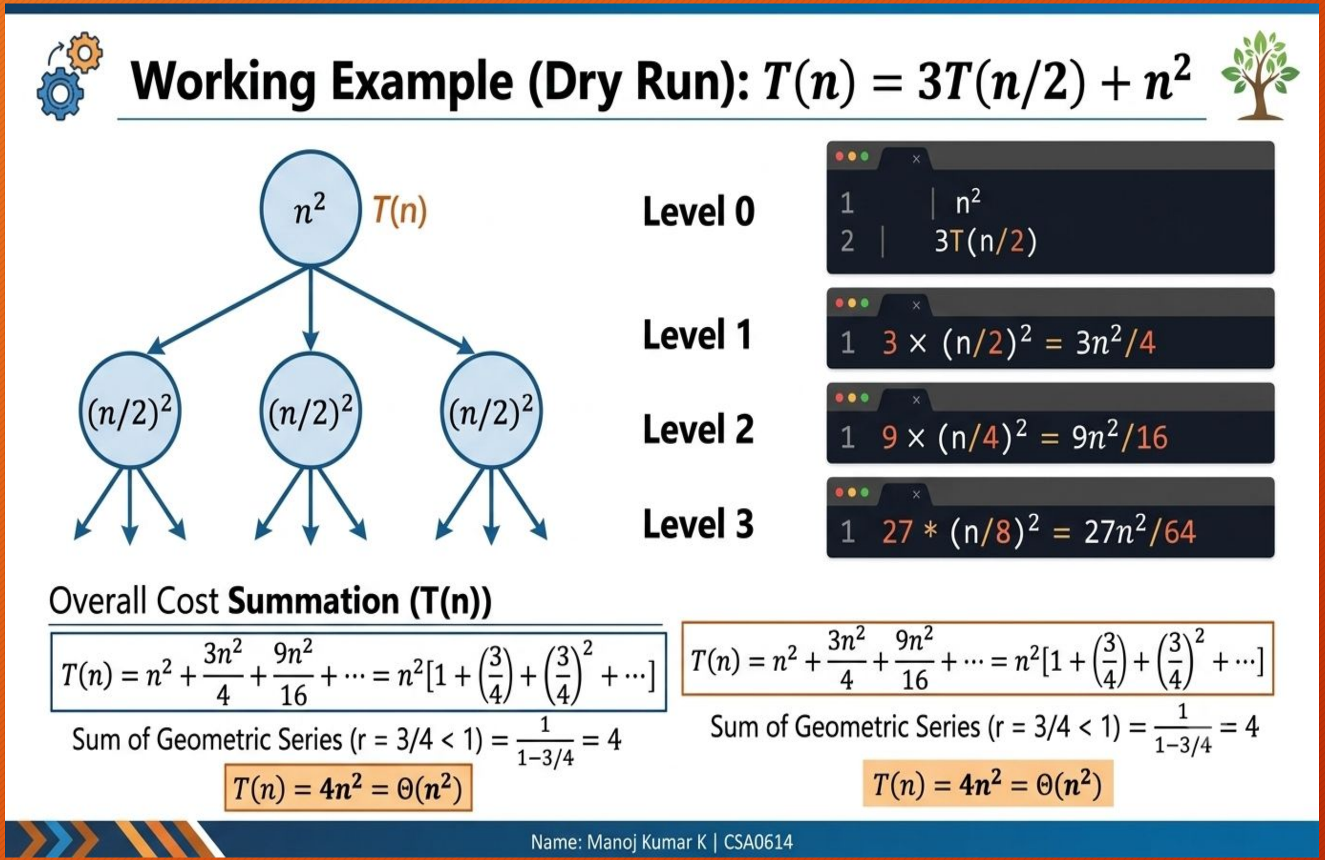
$\mathbf{T(n)=3T(n/2)+n^2}$

- Branches = 3
- Subproblem size = $n/2$
- Extra work = n^2

Algorithm / Pseudocode

```
Algorithm: Mixed Work(n)

Input: Integer n
Output: Total work T(n)

1. If n <= 1
      return 1

2. x1 = Mixed Work(n/2)
3. x2 = Mixed Work(n/2)
4. x3 = Mixed Work(n/2)

5. Perform n × n units of work

6. return x1 + x2 + x3 + n²
```

Working Example (Dry Run)



General level:

$$Cost(i) = (3/4)^i \times n^2$$

Height of tree:

$$\log_2 n$$

Complexity Analysis / Comparative Analysis

Complexity Analysis:
- The recurrence relation is $T(n) = 3T(n/2) + n^2$.
- In the recursion tree, the cost at each level decreases by a factor of 3/4.
- The costs form a geometric series.
- Since the common ratio (3/4) is less than 1, the series converges . Therefore, the total running time is $\Theta(n^2)$.

Comparison with Master Theorem:
- Here, $a = 3$, $b = 2$, and $f(n) = n^2$.
- Since n^2 grows faster than $n^{\wedge}(\log_2 3) \approx n^{\wedge}1{,}585$, the recurrence satisfies Case 3 of the Master Theorem .
- The Master Theorem also gives the time complexity as $\Theta(n^2)$.

Applications

Applications of the Recursion Tree Method

- Divide and Conquer Algorithms
- Merge Sort Analysis
- Quick Sort Analysis
- Binary Search Analysis
- Algorithm Complexity Analysis
- Dynamic Programming Recurrence Analysis

Results & Discussion

Result

- Using the Recursion Tree Method,
- $T(n)=3T(n/2)+n^2$
- has total complexity
- $\Theta(n^2)$

Discussion

- Work decreases at every level.
- The root contributes the largest cost.
- Lower levels contribute less because of the factor **(3/4)^i.**
- The recursion tree result matches the Master Theorem.

Conclusion & References

Conclusion

- The recursion tree was constructed successfully.
- Cost at every level follows a geometric series.
- Summing all levels gives

$\Theta(n^2)$

- The result is confirmed by the Master Theorem